

PREGUNTAS-TIPO-TEST-ANOS-ANTERIO...



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Materiales Aeroespaciales I



2º Grado en Ingeniería Aeroespacial



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PREGUNTAS TIPO TEST PARCIAL

MATERIALES I

1. Which of the following statements is TRUE
- a. Elastic deformation result in movement of dislocations on the microstructural level
 - b. Plastic deformations result in reversible changes in the interatomic distances
 - c. If the stress values are proportional to strain, the modulus of elasticity is stress-dependent
 - d. **On the microscopic level plastic deformations correspond to irreversible displacements of atoms.**

No

2. A 6400 N force is applied to a 5mm diameter nickel wire having a yield strength of 250MPa and a tensile strength of 310MPa. What will happen to the wire?
- a. The wire will deform plastically and uniformly
 - b. **The wire will fracture**
 - c. **The wire will experience necking**
 - d. The wire will recover its original length when unloaded

No

3. Which strengthening mechanism in metallic materials is based on significant increase in number of dislocations:
- a. Solid solutions hardening
 - b. **Strain hardening**
 - c. Precipitation hardening
 - d. Grain refining

4. With an increase in the temperature the electrical conductivity of metals:
- a. **Decreases due to higher scattering of electrons**
 - b. Increases due to higher number of electrons
 - c. Increases due to higher number of lattice imperfections
 - d. Decreases due to lower thermal vibrations of atoms

No

5. Extrinsic semiconductor behaves as an intrinsic one only:
- a. If it is a p type
 - b. **At very high temperature**
 - c. If it is an n type
 - d. At very low temperatures

No

6. Microstructure of steel with 0.5 wt% C below 727°C consists of:
- a. Pearlite matrix with proeutectoid austenite
 - b. **Pearlite matrix with proeutectoid ferrite**
 - c. Austenite matrix with proeutectoid ferrite
 - d. Austenite matrix with proeutectoid cementite

1	Which one of the following statements is FALSE :
X	Polarity of a molecule depends exclusively on the polarity of the chemical bonds.
	Ionic compounds do not conduct electricity except in solution and molten state.
	Hydrogen acquires a partial positive charge when it bonds with an electronegative atom.
	Materials that present strong atomic bonding have low linear coefficients of thermal expansion.

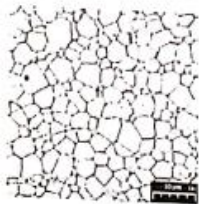
2	A solid substance was tested in the laboratory and the results are listed below. Identify the substance based on them. - It dissolves in water - It is an electrolyte - It melts at a high temperature
	Cu
X	CuBr ₂
	Graphite
	C ₆ H ₁₂ O ₆ (sugar)

3	The plane (1 1 1) is parallel to:
	(1 1 1)
	(1 1 1)
X	(1 1 1)
	(1 1 1)

4	Determine the Miller indices of the line of intersection of (1 1 1) and (1 1 0):
	[1 1 0]
	[1 0 1]
	[1 0 1]
X	[1 1 0]

5	Schottky defect in ionic crystals is a:
	Interstitial impurity
X	Pair of nearby cation and anion vacancies
	Vacancy-interstitial pair of cations
	Substitutional impurity

6	Calculate the activation energy for vacancy formation in aluminum given that the equilibrium number of vacancies at 500 °C is 7.55×10^{23} vacancies/ m^3 . The atomic weight and density (at 500 °C) for aluminum are, respectively, 26.98 g/mol and 2.62 g/ cm^3 . $K = 8.62 \times 10^{-5}$ eV/K; $N_A = 6.022 \times 10^{23}$ atoms/mol.
	1.55 eV/atom
	2.63 eV/atom
X	0.75 eV/atom
	1.29 eV/atom

7	The figure in an ionic solid is a(n):	
X	Grain boundary	
	Edge dislocation	
	Frenkel defect	
	Twin boundary	

8	Which of these variables do not affect diffusion for a case that is described by Fick's First Law?
	Temperature
	Concentration
X	Number of vacancies
X	Time

9	Diffusivity is greater when...
X	solute diffuses interstitially
	temperature is decreased
	crystalline structure is more compact
	number of vacancies is decreased

10	Determine the number of kilograms of hydrogen that pass per hour through a 5 mm thick sheet of palladium having an area of 0.25 m^2 at 500 °C. Assume a diffusion coefficient of 1.0×10^{-8} m^2/s , that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium, and that steady-state conditions have been attained.
X	3.2×10^{-3} kg/h
	0.9×10^{-6} kg/h
	3.2×10^{-6} kg/h
	6.8×10^{-3} kg/h

4,75

2. (5 points) The plane $(1\ 0\ \bar{1})$ of a cubic cell is shown below; the circles represent atoms and they correspond to iron. Atomic linear density for the direction $[1\ 1\ 0]$ is 2.46 atoms/nm.



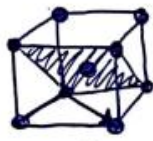
- Indicate the type of crystalline structure, the lattice parameter (a) and the density of the metal in g/cm^3 .
- Calculate the planar density in the plane $(1\ 1\ 0)$ in atoms/ cm^2 .
- After a certain thermal treatment, density of iron results $7.83\ \text{g/cm}^3$. Determine the total number of vacancies in a cubic centimeter of Fe that appear after the increase in temperature.
- Calculate the Burgers vector magnitude for the dislocations movement in the more favourable direction for the Fe at room temperature.

Data: $M(\text{Fe}) = 55.8\ \text{g/mol}$; $N_A = 6.022 \times 10^{23}\ \text{atoms/mol}$.

a)

$M(\text{Fe}) = 55.8\ \text{g/mol}$; $N_A = 6.022 \times 10^{23}\ \text{atoms/mol}$.

a)



Body centered cubic ✓

$$\rho_{[110]} = \frac{1}{L} = 2.46\ \text{atoms/nm} \quad L = 0.4065\ \text{nm}$$



$$a = \frac{4R}{\sqrt{3}}$$

$$\sqrt{2}a = L$$

$$a = 0.2874\ \text{nm}$$

$$\rho_v = \frac{M \cdot n}{N_A \cdot V} = \frac{55.8\ \text{g/mol} \cdot 2}{6.022 \cdot 10^{23} \cdot (0.2874 \cdot 10^{-7})^3} = 7.181\ \text{g/cm}^3$$

$$= 7.181\ \text{g/cm}^3$$

1,25

b)

$$\rho_{(110)} = \frac{2}{\sqrt{2}a \cdot a}$$

$$= 1.71 \cdot 10^{15}\ \text{atoms/cm}^2 \sim 171\ \text{atoms/nm}^2$$

1,25



c)

$$7.181\ \text{g/cm}^3 \rightarrow 7.83\ \text{g/cm}^3$$

1,0

c) $\sqrt{2}a$

$7181 \text{ g/cm}^3 \rightarrow 7183 \text{ g/cm}^3$ 1,0

$7183 \text{ g/cm}^3 = \frac{5518(1-x) \text{ g/mol} \cdot \text{at}}{6.022 \cdot 10^{23} \text{ at/mol} \cdot (0.2874 \cdot 10^{-7})^3 \text{ cm}^3}$ $x = -0.006 ??$

Vacancies per unit cell

$0.006 \text{ vac/cell} \cdot \frac{1 \text{ cell}}{a^3 \text{ cm}^3} =$

$= \frac{2.53 \cdot 10^{20} \text{ vac}}{\text{cm}^3}$

d)  $\langle 111 \rangle$

MORE COMPACT DIRECTION (CUBE DIAGONALS)

$D = 4R = 2|\bar{b}|$

$|\bar{b}| = 2R = \frac{\sqrt{3}a}{2} = 0.2489 \text{ nm}$ 1,25

$R = \frac{\sqrt{3}a}{4}$

1. Two different type of elements are required in order to produce:

- Ionic bonding**
- Covalent bonding
- Metallic bonding
- High thermal fluctuating dipoles

2. In materials with ionic bonding:

- When the ion size increases, the lattice energy increases.
- When the lattice energy is high, usually the melting point is high.**
- The melting temperature usually decreases when the valence of ions increases.
- When the electronegativity difference of the ions is low, the percentage of covalent character of the bond is low.

3. In polymeric materials:

- Bonding between the atoms of molecules within the chains is primarily due to non-directional covalent bonds.
- Bonding in-between the chain is due to covalent bonds.
- Bonding between the atoms of molecules within the chains is particularly directional with a fixed angle between the bonds.**
- Bonding between the atoms of molecules within the chains is primarily due van der waals bonds.**

4. The lattice energy in kJ/mol of the following oxides is:

MgO: 3932; CaO: 3583; SrO: 3311; BaO: 3127. Which of the oxides will most likely have a higher coefficient of thermal expansion?

- a. SrO
- b. CaO
- c. BaO
- d. MgO

5. Which of the following statements is FALSE:

- a. C diffuses by interstitial diffusion in FCC Iron
- b. Nickel diffuses by vacancy diffusion in Copper
- c. Fe diffuses by self-diffusion in BCC Iron
- d. C diffuses by vacancy diffusion in BCC Iron

6. Decreasing the grain size in a material usually:

- a. Increases the strength of a material
- b. Increases the ductility of a material
- c. Increases the melting temperature of a material
- d. Increases the dislocation density of the material

7. Diffusivity of an element:

- a. Depends linearly on temperature
- b. Does not depend on the temperature, but it depends on error function
- c. Depends exponentially on temperature
- d. Does not depend neither on temperature, nor the error function

Element	Atomic radius (mm)	Crystal structure	Electronegativity	Valence
Cu	0.128	FCC	1.8	+2
Fe	0.124	BCC	1.7	+2 +3
Zn	0.133	HCP	1.7	+2
Ni	0.125	FCC	1.8	+2

8. Using the data below order the following elements with respect to their relative degree of atomic solubility in Cu:

- a. Ni>Zn>Fe
- b. Ni>Fe>Zn
- c. Zn>Fe>Ni
- d. Fe>Ni>Zn

9. Diffusion occurs faster:

- a. In materials with high melting point
- b. By vacancy diffusion than by interstitial diffusion
- c. In the BCC structure rather than in the FCC structure
- d. Along the volume (crystal lattice) than along the grain boundary

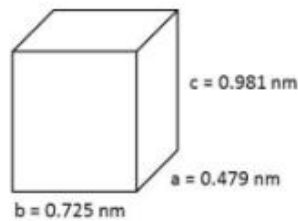
10. The burgers vector in FCC metals is a close packed direction of the form:

- a. $\langle 100 \rangle$
- b. $\langle 0001 \rangle$
- c. $\langle 111 \rangle$
- d. $\langle 110 \rangle$

3. (1.5 points) Iodine has a base centered orthorhombic crystal structure for which the lattice parameters are shown below.

- a. (0.5 p) If the atomic packing factor is 0.547, determine the number of atoms in each unit cell.
- b. (0.25 p) The atomic weight of iodine is 126.91 g/mol; compute its theoretical density.
- c. (0.25 p) Calculate the planar density in the plane (110) in atoms/cm².
- d. (0.5 p) Determine the activation energy for vacancy formation in iodine, given that the equilibrium number of vacancies at 650 °C is 7.68×10^{18} vacancies/cm³.

Data: $R = 8.314 \text{ J/mol K}$.



Solution:

a) 4.3 atoms

b) 2.66 g/cm^3

c) $2.35 \times 10^{14} \frac{\text{atoms}}{\text{cm}^2}$

d) 56.81 kJ/mol

1. In materials with metallic bonding the higher the number of valence electrons involved, the lower the bonding energy and melting point.

F

2. Acetylene ($\text{HC}\equiv\text{CH}$) presents a higher C-C bond energy than that of the ethylene ($\text{H}_2\text{C}=\text{CH}_2$).

T

3. When the ion size in materials with ionic bonding increases the lattice energy decreases.

T

4. Materials with a strong atomic bonding present a low Young's modulus.

F

2. (1 point) BCC lithium has a lattice parameter of $3.5089 \text{ \AA}$ and contains one vacancy per 200 unit cells. Calculate:

- (0.5p) The number of vacancies per cubic centimeter.
- (0.5p) The density of Li.

Data: $M_{\text{Li}} = 6.94 \text{ g/mol}$; $N_A = 6.023 \cdot 10^{23} \text{ atoms/mol}$.

a) $1.157 \cdot 10^{20} \text{ vacancies/cm}^3$

b) 0.532 g/cm^3

1. An ionic compound does not conduct electricity in the solid state but it when dissolved in water the resulting solution conducts electricity.

T

2. Metals form molecular compounds.

F

3. Potassium presents a higher bonding energy than calcium.

F

4. Materials that exhibit high linear coefficient of thermal expansion have a strong atomic bonding and they are useful in applications where the components repeatedly undergo rapid heating and cooling.

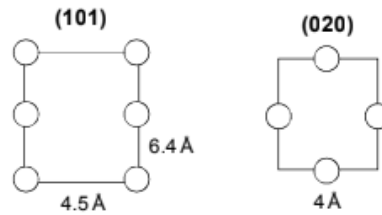
F

3. (2 points) Two distinct crystallographic planes in the unit cell of a hypothetical metal are shown below. The circles represent atoms.

- (0.5 p) Identify the type of crystalline structure and the packing factor.
- (0.5 p) Calculate the planar density for the plane (111) in atoms/cm².
- (0.5 p) Given that the density of the material is $\rho = 6.85 \text{ g/cm}^3$, calculate the molecular weight of the metal.

At a certain temperature a polymorphic change takes place and crystalline structure is transformed to a FCC structure.

- (0.5 p) Determine the change in the volume of the metal after that transformation (expressed in %).



Solution:

- 0.76
- $1.14 \cdot 10^{15} \text{ atoms/cm}^2$
- 92.9 g/mol
- Increases ~3%
